Arabic Cyberbullying Detection Using Deep Learning

A Deep Learning Project Using LSTM, Weighted LSTM, and GRU

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Introduction

Cyberbullying has become a serious problem on social media, especially when people use harmful language to attack or insult others online. In Arabic online communication, this problem is also important because of the large number of users and the different writing styles that appear in tweets and comments. Because of this, it is useful to build automatic systems that can detect bullying text and help make online spaces safer.

In this project, I worked on **Arabic cyberbullying detection** using deep learning. The main goal was to classify Arabic tweets into two classes, bullying and non-bullying. I used the **ArbCyD** dataset, which contains labeled Arabic tweets, and I tested different recurrent deep learning models such as LSTM and GRU. At the end, I compared the models and selected the one that gave the best results for detecting bullying tweets.

Problem Statement

The main problem in this project is how to detect cyberbullying automatically in Arabic tweets. This is important because social media contains a huge number of comments and posts every day, so it is not easy to check all of them manually. Harmful language can affect people badly, especially younger users, and it can also create unsafe online environments.

The challenge in this task is that Arabic text is not always simple to process. People may write in different ways, use dialects, misspell words, or write in informal style. Because of this, detecting bullying in Arabic text is not an easy task. In this project, the aim was to build a deep learning model that can classify tweets into bullying or non-bullying classes as correctly as possible.

Dataset Description

For this project, I used the **ArbCyD dataset**, which is an Arabic cyberbullying dataset collected from social media text. The dataset file used in the notebook was called **ArbCyD Dataset.csv**. It contains **10,000 rows** and **3 columns**, which are **Tweets, Domain, and Label**.

The Tweets column contains the Arabic tweet text, the **Domain column** gives extra category information, and the **Label column** shows whether the tweet is **bullying** or **non-bullying**. The dataset did not contain missing values, which made it easier to work with. In the label distribution, there were 6,204 non-bullying tweets and 3,796 bullying tweets. This means the dataset was slightly imbalanced, so this issue had to be considered later during model training and evaluation.

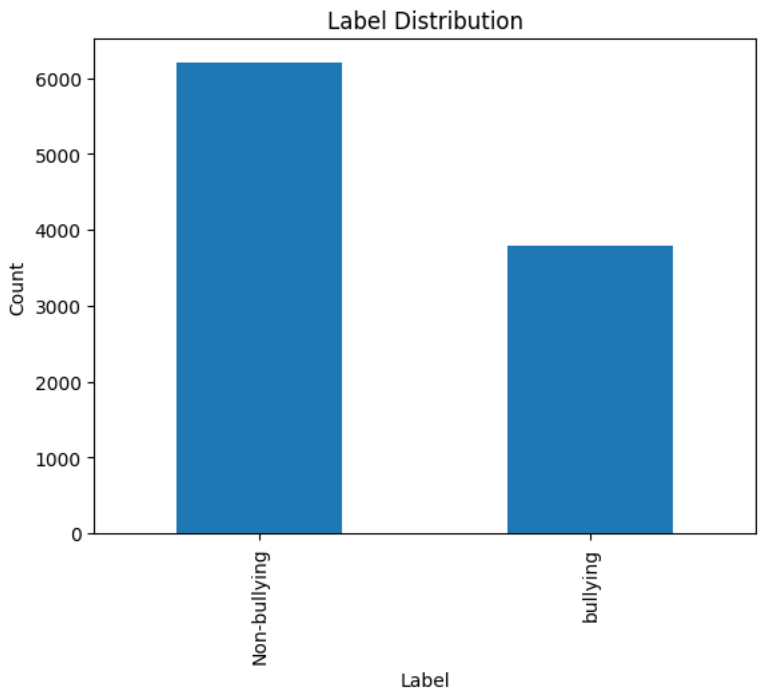

Figure 1. Label distribution of bullying and non-bullying tweets.

Data Preprocessing and Exploration

Before training the models, I explored the dataset to understand it better. The dataset had **10,000 tweets** and no missing values, which was helpful because it meant the data was already clean in terms of structure. I also checked the class labels and found that the dataset contained two classes**, bullying and non-bullying**. The non-bullying class had more examples, so the data was a little imbalanced.

I also looked at tweet length to understand how short or long the tweets were. The shortest tweet had 6 characters, the longest tweet had 281 characters, and the average tweet length was around 121 characters. This gave a better idea about the text size before tokenization. The graphs also showed that bullying and non-bullying tweets had similar length distributions, which means tweet length alone was not enough to separate the classes.

After that, I prepared the data for modeling. The labels were converted into numbers, with **non-bullying as 0 and bullying as 1**. Then the data was divided into training, validation, and test sets using stratified splitting. This was important because it kept the label ratio almost the same in all sets. Finally, the tweets were tokenized and padded so that each input sequence had the same length, which was **30 tokens** in this project.

Methodology

The work in this project started with loading and exploring the dataset to understand its structure, labels, and text characteristics. First, I checked the dataset shape, column names, data types, and missing values. After that, I explored the label distribution and found that the dataset was slightly imbalanced, with more non-bullying tweets than bullying tweets. I also analyzed tweet lengths to better understand the text before building the models.

Next, I prepared the data for deep learning. The labels were encoded into numeric values, where non-bullying was represented as 0 and bullying as 1. Then the dataset was split into **training, validation, and test sets** using stratified splitting, so that the class distribution stayed almost the same in all subsets. After that, **tokenization and padding** were applied to convert the Arabic tweets into numerical sequences of equal length that could be used as input for the models.

For the modeling part, I trained and compared three recurrent deep learning models. The first model was a baseline LSTM. The second model was a weighted LSTM, where class weights were used to reduce the effect of class imbalance. The third model was a weighted GRU model. Early stopping was also used during training to reduce overfitting and restore the best model weights. Finally, the models were evaluated **using test accuracy, confusion matrix, precision, recall, and F1-score.**

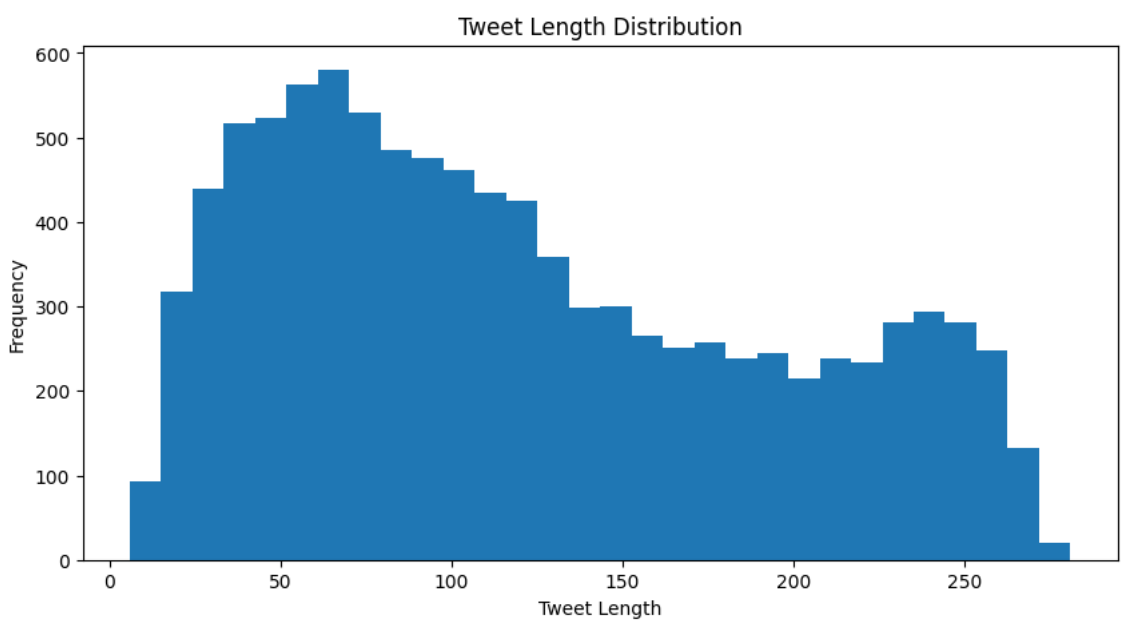

Figure 2. Distribution of tweet lengths in the dataset.

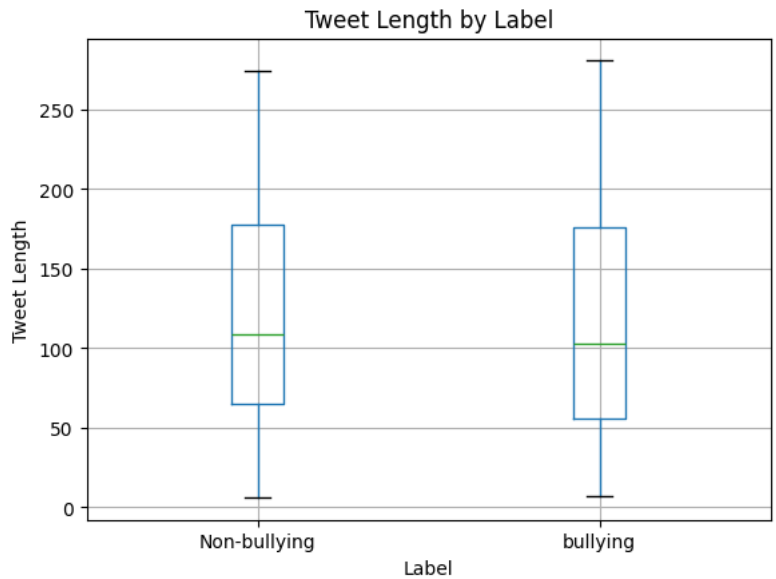

Figure 3. Tweet length comparison between bullying and non-bullying tweets.

Models and Training Process

In this project, I **tested three deep learning models** for Arabic cyberbullying detection. The first model was a **baseline LSTM model**. It included an **embedding layer**, **one LSTM layer**, a **dropout layer,** and a **final dense output layer with sigmoid activation for binary classification**. This model was used as the starting point to see how well a simple recurrent model could perform on the dataset.

After that, I trained a second model using the same **LSTM architecture** but with **class weights**. The reason for this was that the dataset had more non-bullying tweets than bullying tweets, so the model needed to pay more attention to the bullying class. Using class weights helped reduce the effect of imbalance and improved the model's ability to detect harmful tweets.

The **third model** was a **GRU model with class weights**. This model had a similar structure to the LSTM model, but it used a **GRU layer instead** of an LSTM layer. I tried this because GRU is also a recurrent deep learning model for sequence data and it was covered in the course. All models were trained using the **Adam optimizer and binary crossentropy** loss. Early stopping was also used during training to stop the process when validation loss stopped improving and to restore the best model weights.

Results and Discussion

The baseline LSTM model gave a test accuracy of about 76.6%. It performed reasonably well on both classes, but its recall for the bullying class was still lower than desired. This means that the model was able to classify many tweets correctly, but it still missed a noticeable number of actual bullying tweets. The bullying recall in this model was 0.61, which showed that the baseline model learned useful patterns but was not the strongest option for the main goal of the project.

To improve this issue, I trained a weighted LSTM model using class weights. This model gave the best overall result, with a test accuracy of about 77.1%. It also improved the bullying recall to 0.69 and achieved the best bullying F1-score of 0.70. This was an important improvement because it showed that the model became better at identifying bullying tweets while still keeping a good overall balance in performance. The final weighted LSTM model also produced a confusion matrix of [[762, 169], [175, 394]], which shows that it detected more bullying tweets correctly than the baseline model and achieved the best overall balance in the final results.

I also tested a weighted GRU model for comparison. This model gave a test accuracy of about 76.2%. Its bullying recall was 0.57 and its bullying F1-score was 0.64. Although the weighted GRU still gave acceptable results, it did not perform better than the weighted LSTM in the final test results. In general, the results show that handling class imbalance was important in this project, and that the weighted LSTM was the most suitable final model for this dataset.

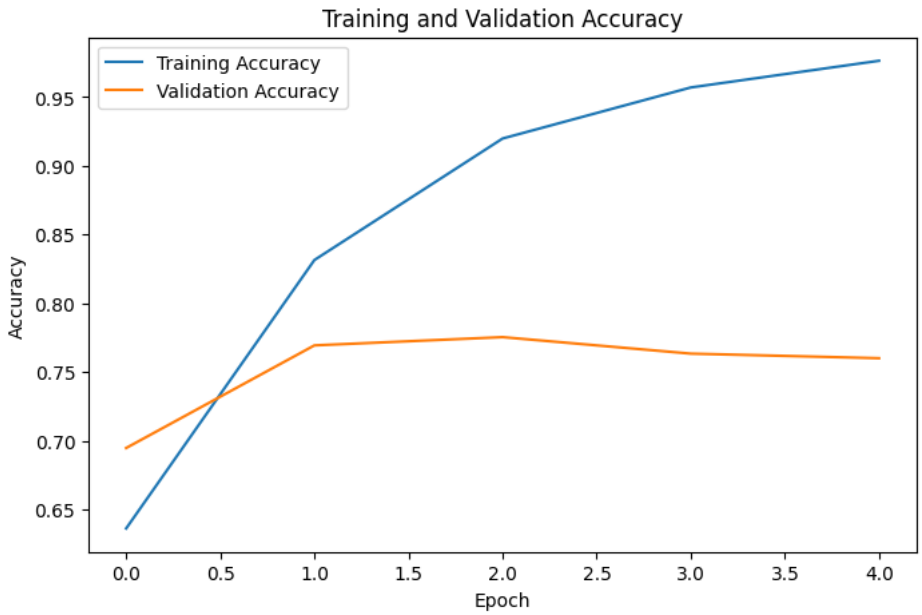

Figure 4. Training and validation accuracy of the baseline LSTM model.

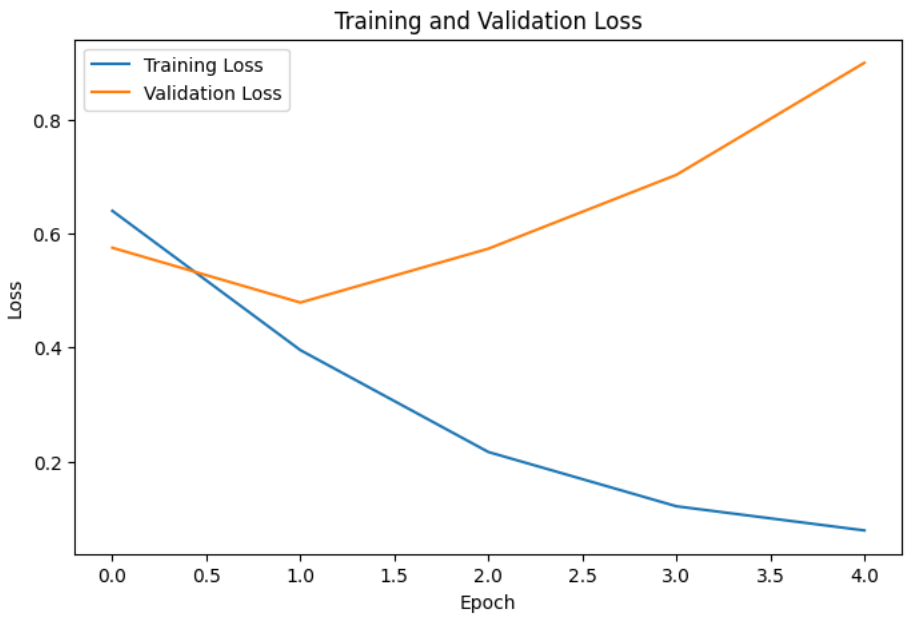

Figure 5. Training and validation loss of the baseline LSTM model.

	Model	Test Accuracy	Bullying Precision	Bullying Recall	Bullying F1-score
0	Baseline LSTM	0.7660	0.73	0.61	0.67
1	Weighted LSTM	0.7707	0.70	0.69	0.70
2	Weighted GRU	0.7620	0.75	0.57	0.64

Table 1. Comparison of the main results of all tested models.

Based on the comparison table, the weighted LSTM model was selected as the final model for this project. Although all three models achieved similar overall accuracy, the weighted LSTM gave the best balance between general performance and bullying detection in the final results. It achieved the highest test accuracy among the tested models and also produced the strongest bullying recall and F1-score. This means it was better at identifying harmful tweets than the baseline LSTM and the weighted GRU. Since the main objective of the project is to detect cyberbullying effectively, the weighted LSTM was considered the most suitable final model.

Conclusion and Future Work

In this project, I developed a deep learning approach for Arabic cyberbullying detection using the ArbCyD dataset. The work started with dataset exploration and preprocessing, followed by training and comparing different recurrent deep learning models. The results showed that a simple baseline LSTM model was able to learn useful patterns, but it was not the strongest model for detecting bullying tweets. After that, using class weights improved the performance, and the weighted LSTM model gave the best final result among all tested models.

The final weighted LSTM model achieved the highest test accuracy and also gave the best results for the bullying class, especially in recall and F1-score. This means it was better at identifying harmful tweets and more suitable for the main goal of the project. Overall, this project showed that recurrent deep learning models can be effective for Arabic text classification, especially when the data is prepared carefully and class imbalance is handled during training.

For future work, this project could be improved by trying pretrained Arabic transformer models, testing different hyperparameters, or applying more advanced Arabic text preprocessing techniques. It may also be useful to test the models on other Arabic datasets to see if the performance stays strong on different types of social media text.

References

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